

Operating systems

Memory

Content Lecture#13

01	M
02	A
03	S
04	P
05	A

Lets take a Break

A programmer had a problem, so he used threds. Then he had two problems.



I went to an I.T.-themed restaurant the other day. It had motherboards on the walls, the placemats looked like keyboards, the cutlery had USB sticks for handles, you get the idea. But the waitstaff seemed sad. Really, really sad. The host was sighing as we walked to my The bartender didn’t like the space bar. And I caught my waiter crying after he took my order. I didn’t understand why all the servers were down.

So when I saw the task manager, I asked him what was wrong. He said, seems your food keeps getting stuck in testing. I asked if it needed to be debugged. He told me those aren’t bugs, they’re features. I asked if I could get some chips while I waited. He told me only if I also accepted all cookies. I said no. Then BLAM! He hit me. I said what the hell was that?! He said, a denial of service attack. I started to curse him out, but he threw me out for being a cursor. Jokes on him though: I didn't have enough cache anyway.



Virtual Memory

A

Address binding

is

Memory

Q: Calculate the number of bits required in the address for memory having size of 16 GB. Assume the memory is 4-byte addressable.

Memory

Solution

Let ‘n’ number of bits are required. Then,

Size of memory = 2^n x 4 bytes.

Since, the given memory has size of 16 GB, so we have-

2^n x 4 bytes = 16 GB

2^n x 4 = 16 G

2^n x 2^2 = 2^{34}

2^n = 2^{32}

∴ n = 32 bits

Memory

Q: Consider a system with byte-addressable memory, 32 bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. The size of the page table in the system in megabytes is _____.

Memory

Given-
•Number of bits in logical address = 32 bits
•Page size = 4KB
•Page table entry size = 4 bytes

Process Size-

Number of bits in logical address = 32 bits
Thus,
Process size
= 2^{32} B
= 4 GB

Number of Entries in Page Table-

Number of pages the process is divided
= Process size / Page size
= 4 GB / 4 KB
= 2^{20} pages

Thus,
Number of entries in page table = 2^{20} entries

Page Table Size-

Page table size
= Number of entries in page table x Page table entry size
= 2^{20} x 4 bytes
= 4 MB is the answer

Memory

Q: Consider a machine with 64 MB physical memory and a 32 bit virtual address space. If the page size is 4 KB, what is the approximate size of the page table?

1. 16 MB
2. 8 MB
3. 2 MB
4. 24 MB

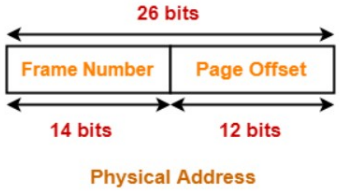
Memory

•Size of main memory = 64 MB
•Number of bits in virtual address space = 32 bits
•Page size = 4 KB
We will consider that the memory is byte addressable.

Number of Bits in Physical Address-
Size of main memory
= 64 MB
= 2^{26} B
Thus, Number of bits in physical address = 26 bits

Number of Frames in Main Memory-
Number of frames in main memory
= Size of main memory / Frame size
= 64 MB / 4 KB
= 2^{26} B / 2^{12} B
= 2^{14}
Thus, Number of bits in frame number = 14 bits

Number of Bits in Page Offset-
We have,
Page size
= 4 KB= 2^{12} B Thus, Number of bits in page offset = 12 bits
So, Physical address is-



rocess Size-

Number of bits in virtual address space = 32 bits
Thus,
Process size
= 2^{32} B
= 4 GB

Number of Entries in Page Table-

Number of pages the process is divided
= Process size / Page size
= 4 GB / 4 KB
= 2^{20} pages

Thus, Number of entries in page table = 2^{20} entries

Page Table Size-

Page table size
= Number of entries in page table x Page table entry size
= Number of entries in page table x Number of bits in frame number
= 2^{20} x 14 bits
= 2^{20} x 16 bits (Approximating 14 bits ≈ 16 bits)
= 2^{20} x 2 bytes
= 2 MB Answer

Page replacement

Suppose the CPU generates the following sequence of 15 logical addresses while executing a process P. [1010, 0413, 0691, 0010, 1020, 0575, 1024, 0500, 0780, 0220, 0340, 0381, 0268, 0950, 0321] i. Calculate the number of page faults using ‘Least Frequently Used’ page replacement strategy, if page-size = 200 bytes and no. of frames allocated to process P is 4 (Assume that word-size = 1 byte).

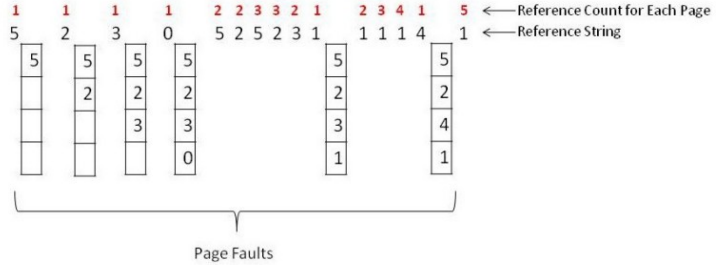
Solution

Given: Page Size = 200 bytes No. of Frames allocated to process P = 4 Word size = 1 byte CPU generated string of logical addresses: [1010, 0413, 0691, 0010, 1020, 0575, 1024, 0500, 0780, 0220, 0340, 0381, 0268, 0950, 0321]

Since, each data word is 1 byte long and page-size is 200 bytes, logical address locations 0 to 199 belong to Page Number 0. Similarly, the mapping for all logical addresses to their corresponding page no.s can be shown using the following table:-

Hence, the CPU generated string of page no.s (Reference String) obtained from the given string of logical addresses is as shown below:- [5, 2, 3, 0, 5, 2, 5, 2, 3, 1, 1, 1, 1, 4, 1] (Note: In this problem, we should not merge the same consecutive page no.s, because in Least Frequently used page replacement approach, the 'Count' of references to each page is important !) Now, the status of the 4 Frames after Every Page Reference is shown below (note that, whenever a page-replacement is required, we are replacing the page with the least reference-count value):-

Adress Location	Page Number
0 – 199 0	0
200 – 399 1	1
400 – 599 2	2
600 – 799 3	3
800 – 999 4	4
1000 - 1199 5	5



Hence, number of page faults = 6

Critical Section

A critical section is a section of code that needs to be executed without outside interference - i.e. without another thread potentially affecting/being affected by "intermediate" states within the section.

For instance, a reservation system might have a critical section when reserving something in that it needs to both check to see if the item is available and then mark it as no longer available, without some other attempt at reserving the room changing that status in the middle.
Thus, the critical section of a piece of code is a place where only one thread of execution is allowed to be at a time, to prevent things like race conditions.

Critical Section

A **critical section** is a sequence of instructions that must be executed **atomically**. That is, a critical section contains multiple instructions that create race conditions if they are **interleaved** with other threads. Every access to a global variable in a multithreaded program creates a critical section.

Exam

The final-term theory exam of OS will consist of four questions, one from each of the topics listed below.

Memory: segmentation, paging, TLB, address translation

Threads

Synchronization

Virtual Memory



Thank you